

Lecture FYS5419, April 24, 2024

FYS 5419/9419, APRIL 24, 2024

Shor's algorithm (1994)

- (i) classical part, grounded in number theory, relies on modulus arithmetic and a process called order finding
- (ii) Classically almost intractable, but can be efficiently implemented with a quantum algo.

$$a = b \underbrace{\text{MOD } N}_{\%}$$

$$1 = 9 \% 4$$

Modulus is the remainder of the Euclidean division

$$a \% b = a - \text{floor}(a/b) \times b$$

$\text{floor}(x)$ = largest integer not greater than x

$$5 \% 7 = 5$$

$$-9 \% 12 = 15$$

$$\text{floor}(-9/12) = -\underline{1}$$

$$(xy) \% (z) = (x \% z)(y \% z)$$

Greatest common divisor,

$$21 = 3 \cdot 7$$

$$N = p_0^{e_0} p_1^{e_1} p_2^{e_2} \dots p_{N-1}^{e_{N-1}}$$

$$32 = 4 \cdot 8 = 2^5 \quad p_0 = 2 \quad e_0 = 5$$

$$99 = 9 \cdot 11 = 3^2 \cdot 11^1$$

$$P_0 = 3 \quad P_1 = 11$$

The most difficult case is when N has only two prime numbers

$$N = P \cdot q$$

The problem of factoring N (large number) into two primes, is equiva-

lent to solving

$$x^2 = 1 \pmod{N}$$

positive integers x and,
 $x < N$, with no common
factors. The order $-r-$
is defined to be the least
positive integer $-r-$, which
solves $x^r = 1 \pmod{N}$

Order finding is believed to be a classically difficult problem.

$$N = 21 = 3 \cdot 7$$

$$1 \times 1 = 1 = 1\% N$$

$$2 \times 2 = 4 = 4\% N \quad 8 \times 8 = 64$$

$$3 \times 3 = 9 = 9\% N = 1\% N$$

$$4 \times 4 = 16 = 16\% N$$

$$5 \times 5 = 25 = 4\% N$$

(Hundt, sec 6.5)

Exercise: show that the order
— 1 — of $x = 5 \bmod 21$ is
6.

The quantum algorithm for
order finding is the
QPE algorithm applied
to the unitary operation

$$U|y\rangle = |xy \bmod N\rangle$$

with $y \in \{0, 1\}^L$

when $N \leq y \leq 2^{L-1}$ we
 use that $x^y \% N$ is
 just y , and u
 acts non-trivially when
 $0 \leq y \leq N-1$

$$|u_s\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \exp\left[\frac{-2\pi i s k}{N}\right] \times |x^{k \% N}\rangle$$

$|001\rangle$ map as $2^0 \cdot 0 + 2^1 \cdot 0 + 2^2 \cdot 1$
 $\uparrow \Rightarrow |4\rangle$ ($\#$ realizations 2^n)

$|g\rangle$

$$u^k |g\rangle = |x^k g \bmod N\rangle$$

